

Fix Report

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Repository: github.com/onestrange-guy/siwes-server-lab

1. Root Cause Restatement (from Subtask 2.2.4)

The static IP addresses required by the lab (10.10.10.10 on DC01, 10.10.10.20 on dns01) were never applied to the correct network adapters on either virtual machine. And both servers have 192.168.100.0/24 subnet. My diagnosis predicted a fix: assign the correct static IPs

What the diagnosis missed: During execution, a deeper issue was discovered; the VirtualBox adapter configuration requires I little adjustment. Ubuntu's Adapter 2 was set to NAT instead of Host-Only. As a result, even when 10.10.10.20/24 was correctly configured, it existed on a virtual NIC that was physically isolated from DC01's Host-Only adapter.

Additionally, Windows Server's static IP 10.10.10.10 was applied to the NAT adapter instead of the Host-Only adapter. Both errors together meant no packet on the 10.10.10.0/24 network could ever traverse the virtual wire between the two VMs.

Before making changes, the diagnosis from 2.2.4 was reviewed against the live state.

Predicted Fix	Still Valid?	Note
Apply 10.10.10.10 to Windows Host-Only adapter	Yes	IP was missing entirely
Apply 10.10.10.20 to Ubuntu Host-Only adapter	Yes	IP was missing entirely

Due to the missing IP addresses on both servers, I want to reconfigure the IP addresses to a static IP to match 10.10.10.0/24.

Command Execution Log

Windows IP Fix

Server	Command	Expected	Actual
DC01	Get-NetAdapter Format-Table Name, InterfaceDescription, Status, MacAddress	List of adapters with MACs	Ethernet (08-00-27-C7-CE-73) and Ethernet 2 (08-00-27-9A-FD-B7)

DC01	ping 10.10.10.20	Diagnose connectivity state	"Destination host unreachable" from 10.10.10.10
DC01	Get-NetIPAddress - AddressFamily IPv4 Select InterfaceAlias, IPAddress, PrefixLength	See current IP state	Ethernet 2 = 169.254.220.119 (APIPA), Ethernet = 10.0.3.15

Finding: The static IP had not persisted. Ethernet 2 showed an APIPA address (169.254.x.x), indicating no IP configuration at all. Adapter role mapping was also unclear at this point.

Windows IP Applied to Wrong Adapter

Server	Command	Expected	Actual
DC01	Remove-NetIPAddress - InterfaceAlias "Ethernet" - Confirm:\$false	Remove old IP	Executed
DC01	New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 10.10.10.10 - PrefixLength 24	Apply static IP	Applied — but to wrong adapter
DC01	Get-NetIPAddress -AddressFamily IPv4 Select InterfaceAlias, IPAddress, PrefixLength	Verify	Ethernet = 10.10.10.10/24, Ethernet 2 = 169.254.220.119
DC01	ping 10.10.10.20	Replies	"Destination host unreachable" — still failing

Finding: Ethernet (MAC 08-00-27-C7-CE-73) was confirmed to be the NAT adapter, not Host-Only.

DC01 adapter mapping (confirmed via MAC):

VirtualBox	Adapter	MAC	Windows Name	Role
DC01 Adapter 1	Host-Only	08-00-27-9A-FD-B7	Ethernet 2	Host-Only
DC01 Adapter 2	NAT	08-00-27-C7-CE-73	Ethernet	NAT

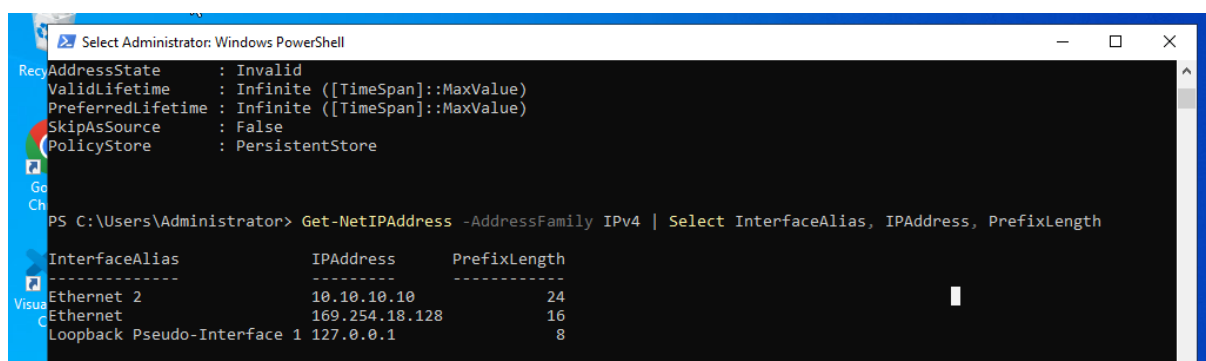
DNS01 adapter mapping (confirmed via MAC):

VirtualBox	Adapter	MAC	Ubuntu Name	Role
DNS01 Adapter 1	NAT	08-00-27-A9-40-64	enp0s3	NAT
DNS01 Adapter 2	NAT	08-00-27-1B-2C-4A	enp0s8	Should be Host-Only

Finding: Ubuntu's Adapter 2 was set to NAT in VirtualBox not Host-Only. This meant enp0s8 (which held 10.10.10.20) was on a completely isolated virtual network that could never communicate with DC01's Host-Only adapter.

Windows: Static IP Reconfiguration

Server	Command	Expected	Actual
DC01	Remove-NetIPAddress -InterfaceAlias "Ethernet" -Confirm:\$false	Remove IP from NAT adapter	Executed
DC01	New-NetIPAddress -InterfaceAlias "Ethernet 2" -IPAddress 10.10.10.10 -PrefixLength 24	Apply to Host-Only adapter	Applied
DC01	Get-NetIPAddress -AddressFamily IPv4 Select InterfaceAlias, IPAddress, PrefixLength	Ethernet 2 = 10.10.10.10/24	Ethernet 2 = 10.10.10.10/24



```
Select Administrator: Windows PowerShell

RecAddressState : Invalid
ValidLifetime   : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource    : False
PolicyStore     : PersistentStore

PS C:\Users\Administrator> Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias, IPAddress, PrefixLength

InterfaceAlias  IPAddress      PrefixLength
-----
Ethernet 2     10.10.10.10    24
Ethernet       169.254.18.128 16
Loopback Pseudo-Interface 1 127.0.0.1      8
```

Ubuntu:

Ubuntu's Netplan configuration already had 10.10.10.20/24 assigned to enp0s8. Once VirtualBox Adapter 2 was changed to Host-Only, no further changes were needed inside the Ubuntu OS. netplan apply had been run in an earlier attempt and the config persisted correctly.

```
siwesadmin@dns01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a9:40:64 brd ff:ff:ff:ff:ff:ff
    altname enx080027a94064
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86359sec preferred_lft 86359sec
    inet6 fd17:625c:f037:2:a00:27ff:fea9:4064/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86361sec preferred_lft 14361sec
    inet6 fe80::a00:27ff:fea9:4064/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1b:2c:4a brd ff:ff:ff:ff:ff:ff
    altname enx0800271b2c4a
    inet 10.10.10.20/24 brd 10.10.10.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe1b:2c4a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
siwesadmin@dns01:~$
```

Errors Encountered and Resolutions

Error	Cause	Resolution
Ping "Destination host unreachable" persisted after OS IP fix	Ubuntu Adapter 2 was NAT wrong VirtualBox wiring	Changed DNS01 Adapter 2 from NAT to Host-Only in VirtualBox
10.10.10.10 disappeared between sessions	IP was applied to NAT adapter which reset	Re-applied to correct adapter (Ethernet 2)
cat /etc/netplan/*.yaml returned "Permission denied"	File owned by root, requires sudo	sudo cat /etc/netplan/*.yaml

Final Verification Results

Ubuntu Server (DNS01) Script Outputs

```
slwesadmin@dns01:~$ cd Document
slwesadmin@dns01:~/Document$ python3 network_info.py
=====
NETWORK INFORMATION REPORT
=====
Script Author : Israel
Date          : 18 June 2026
Python Version: 3.14.4

[SYSTEM]
-----
Hostname       : dns01
Operating System: Linux 7.0.0-22-generic
Detected OS    : LINUX

[IPv4 ADDRESSES]
-----
10.0.2.15
10.10.10.20

[DEFAULT GATEWAY]
-----
10.10.10.1

[DNS SERVERS]
-----
10.0.2.3
8.8.8.8

=====
END OF REPORT
=====
slwesadmin@dns01:~/Document$
```

Windows (DC01) Script Outputs

```
Administrator: Windows PowerShell
Rec PS C:\Users\Administrator> cd .\Documents\
PS C:\Users\Administrator\Documents> cd .\siwes-lab-scripts\
PS C:\Users\Administrator\Documents\siwes-lab-scripts> py network_info.py
=====
NETWORK INFORMATION REPORT
=====
Python Version: 3.14.6

[SYSTEM]
-----
Hostname       : DC01
Operating System: Windows 2022Server
Detected OS    : WINDOWS

[IPv4 ADDRESSES]
-----
10.10.10.10
169.254.18.128

[DEFAULT GATEWAY]
-----
10.10.10.1

[DNS SERVERS]
-----
8.8.4.4
8.8.8.8

=====
END OF REPORT
=====
```

Ping Tests

Windows to Ubuntu:

```
Pinging 10.10.10.20 with 32 bytes of data:
Reply from 10.10.10.20: bytes=32 time=8ms TTL=64
Reply from 10.10.10.20: bytes=32 time=2ms TTL=64
Reply from 10.10.10.20: bytes=32 time=2ms TTL=64
Reply from 10.10.10.20: bytes=32 time=5ms TTL=64

Ping statistics for 10.10.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 8ms, Average = 4ms
PS C:\Users\Administrator>
```

Ubuntu to Windows:

```
silvesadmin@dns01:~$ ping -c 4 10.10.10.10
PING 10.10.10.10 (10.10.10.10) 56(84) bytes of data:
64 bytes from 10.10.10.10: icmp_seq=1 ttl=128 time=3.46 ms
64 bytes from 10.10.10.10: icmp_seq=2 ttl=128 time=2.25 ms
64 bytes from 10.10.10.10: icmp_seq=3 ttl=128 time=2.11 ms
64 bytes from 10.10.10.10: icmp_seq=4 ttl=128 time=3.13 ms

--- 10.10.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 2.107/2.736/3.460/0.571 ms
silvesadmin@dns01:~$ _
```

Verification Item	Command	Expected Result	Actual Result	Pass/Fail
Windows script output	py network_info.py	IP: 10.10.10.10, Gateway: 10.10.10.1	IP: 10.10.10.10, Gateway: 10.10.10.1	Pass
Ubuntu script output	python3 network_info.py	IP: 10.10.10.20, Gateway: 10.10.10.1	IP: 10.10.10.20, Gateway: 10.10.10.1	Pass
Windows → Ubuntu ping	ping 10.10.10.20	Reply received, 0% loss	4/4 replies, avg 4ms	Pass
Ubuntu → Windows ping	ping -c 4 10.10.10.10	Reply received, 0% loss	4/4 replies, avg 2.7ms	Pass